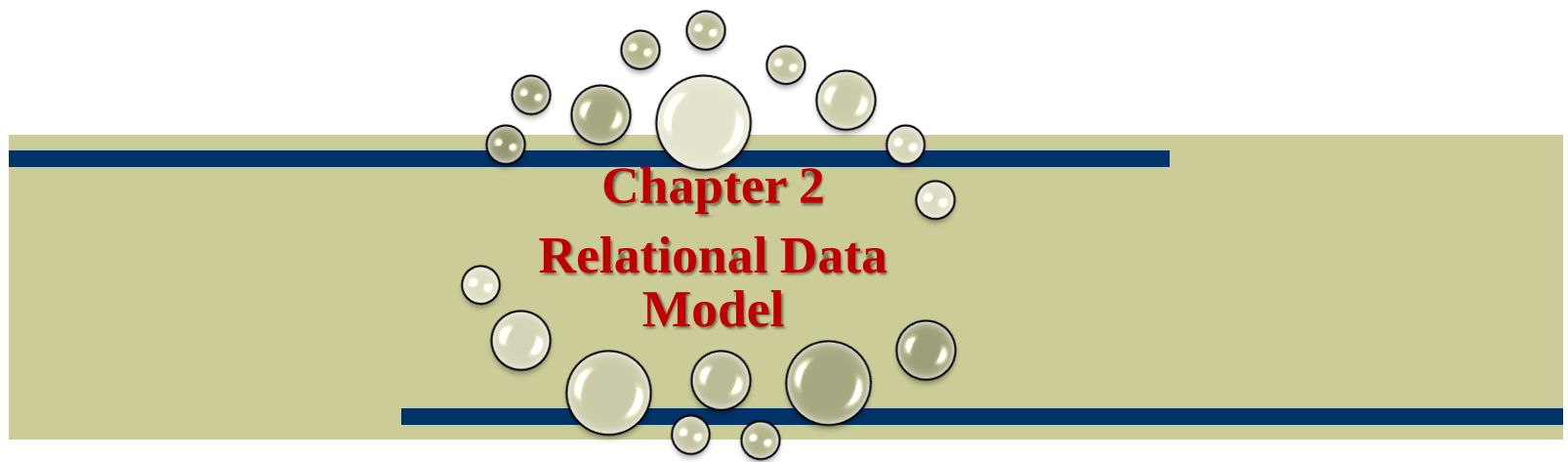




Unit 2.4:
**Introduction to SQL, Data Types in SQL,
Components of SQL – DDL, DML, DCL, DQL.**

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What is SQL?

- ◆ SQL is Structured Query Language
- ◆ A computer language for storing, manipulating and retrieving data stored in relational database.
- ◆ SQL is the standard language for Relation Database System.
- ◆ All RDBMSs like MySQL, MS Access, Oracle, Sybase, Informix and SQL Server use SQL as standard database language.

Why SQL?

- ◆ Allows users to access data in relational database management systems.
- ◆ Allows users to describe the data.
- ◆ Allows users to define the data in database and manipulate that data.
- ◆ Allows to embed within other languages using SQL modules, libraries & pre-compilers.
- ◆ Allows users to create and drop databases and tables.
- ◆ Allows users to create view, stored procedure, functions in a database.
- ◆ Allows users to set permissions on tables, procedures, and views

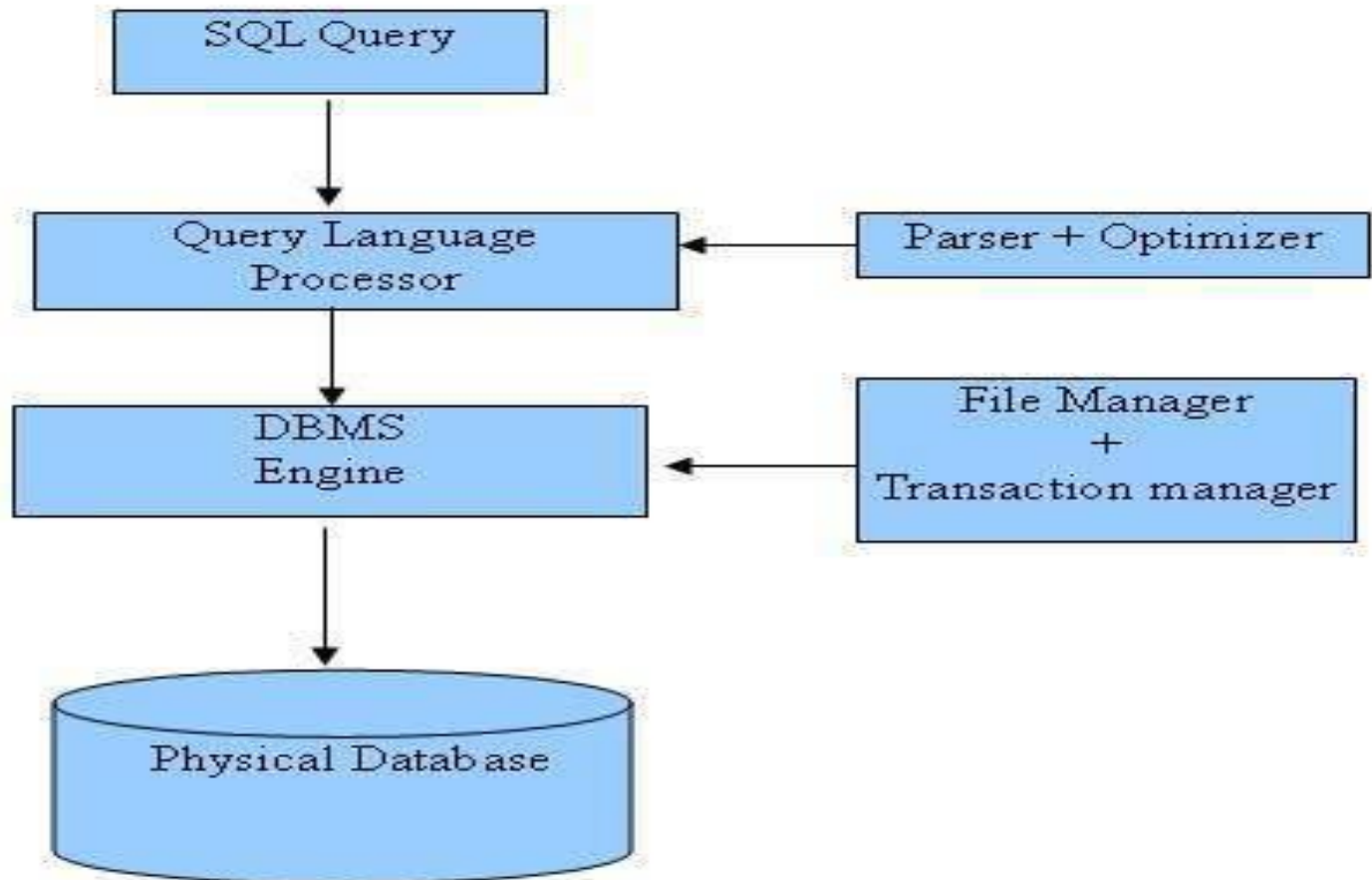
History

- ◆ **1970** -- Dr. Edgar F. Codd of IBM is known as the father of relational databases. He described a relational model for databases.
- ◆ **1974** -- Structured Query Language appeared.
- ◆ **1978** -- IBM worked to develop Codd's ideas and released a product named System/R.
- ◆ **1986** -- IBM developed the first prototype of relational database and standardized by ANSI. The first relational database was released by Relational Software and its later becoming Oracle.

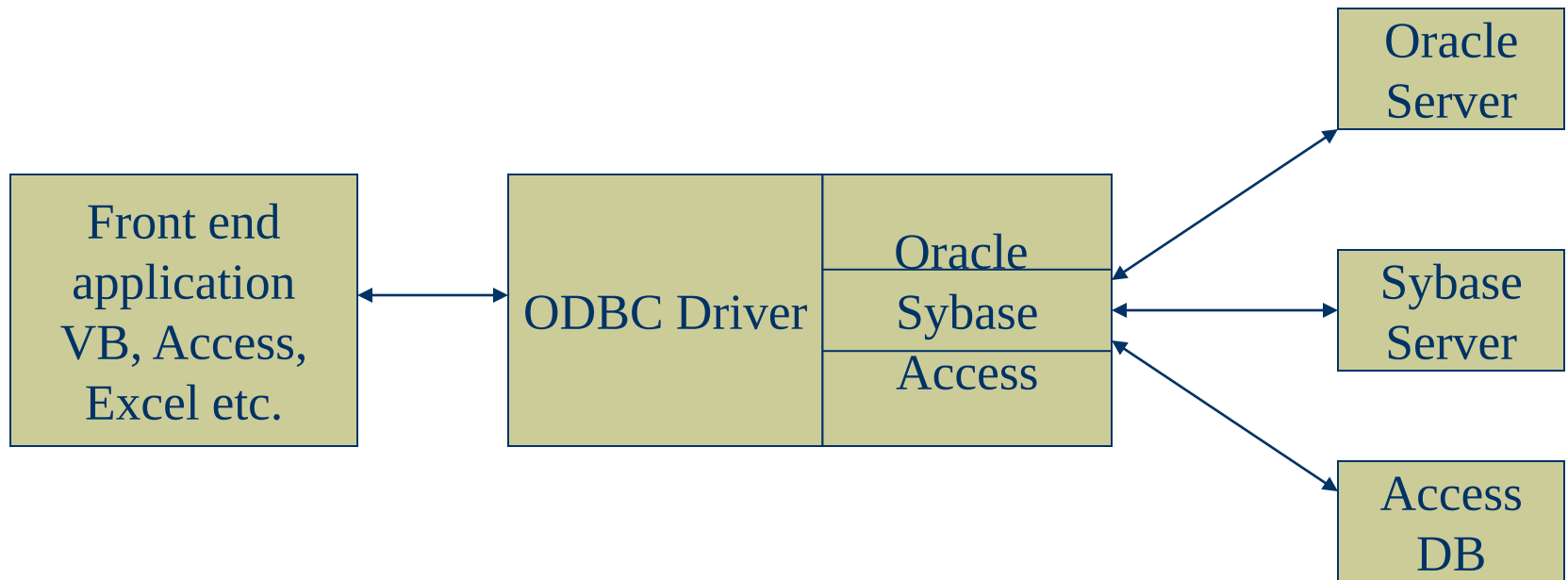
SQL Process:

- ◆ When we are executing an SQL command for any RDBMS, the system determines the best way to carry out your request and SQL engine figures out how to interpret the task.
- ◆ There are various components included in the process. These components are Query Dispatcher, Optimization Engines, Classic Query Engine and SQL Query Engine.

Following is a simple diagram showing SQL Architecture:



Components



Core Database Engine

- **ORACLE RDBMS** (Oracle Universal server)
- **Integrated Data Dictionary**: manage tables owned by all users in a system
- **SQL**: language to access and manipulate data
- **PL/SQL**: a procedural extension to SQL language

Data Types

- ◆ SQL data type is an attribute that specifies type of data of any object.
- ◆ Each column, variable and expression has related data type in SQL.
- ◆ Each column is given a name and a data type that reflects the kind of data it will store.
- ◆ Oracle supports four basic data types
 - CHAR
 - NUMBER
 - DATE
 - RAW.
- ◆ There are also a few additional variations on the RAW and CHAR data types.

◆ VARCHAR2

- Character data type.
- Can contain letters, numbers and punctuation.
- The syntax : VARCHAR2(size) where *size* is the maximum number of alphanumeric characters the column can hold.
- In Oracle, the maximum size of a VARCHAR2 column is 4,000 bytes.

◆ NUMBER

- Numeric data type.
- Can contain integer or floating point numbers only.
- The syntax : NUMBER(precision, scale) where *precision* is the total size of the number including decimal point and *scale* is the number of places to the right of the decimal.
- For example, NUMBER(6,2) can hold a number between -999.99 and 999.99.

♦ **DATE** –

- Date and Time data type.
- Can contain a date and time portion in the format: DD-MON-YY HH:MI:SS.
- No additional information needed when specifying the DATE data type.
- the time of 00:00:00 is used as a default.
- The output format of the date and time can be modified

♦ **RAW** –

- Free form binary data.
- Can contain binary data up to 255 characters.
- Data type LONG RAW can contain up to 2 gigabytes of binary data.
- RAW and LONG RAW data cannot be indexed and can not be displayed or queried in SQL*Plus.
- Only one RAW column is allowed per table.

♦ **LOB** –

- Large Object data types.
- These include BLOB (Binary Large Object) and CLOB (Character Large Object).
- More than one LOB column can appear in a table.
- These data types are the preferred method for storing large objects such as text documents (CLOB), images, or video (BLOB).

Exact Numeric Data Types

DATA TYPE	FROM	TO
bigint	-2^{63} (-9,223,372,036,854,775,808)	2^{63} (9,223,372,036,854,775,807)
int	-2,147,483,648	2,147,483,647
smallint	-32,768	32,767
tinyint	0	255
bit	0	1
decimal	$-10^{38} + 1$	$10^{38} - 1$
numeric	$-10^{38} + 1$	$10^{38} - 1$
money	-922,337,203,685,477.5808	+922,337,203,685,477.5807
smallmoney	-214,748.3648	+214,748.3647

Approximate Numeric Data Types

DATA TYPE	FROM	TO
float	-1.79E + 308	1.79E + 308
real	-3.40E + 38	3.40E + 38

Date and Time Data Types:

DATA TYPE	FROM	TO
DATETIME format: YYYY-MM-DD HH:MI:SS.	Jan 1, 1753	Dec 31, 9999
SMALLDATETIME format: YYYY-MM-DD HH:MI:SS.	Jan 1, 1900	Jun 6, 2079
DATE format YYYY-MM-DD.	Stores a date like June 30, 1991	
TIME format: HH:MI:SS.	Stores a time of day like 12:30 P.M.	

Character Strings Data Types

DATA TYPE	TO
char	Maximum length of 8,000 characters.(Fixed length non-Unicode characters)
varchar	Maximum of 8,000 characters.(Variable-length non-Unicode data).
varchar(max)	Maximum length of 231characters, Variable-length non-Unicode data (SQL Server 2005 only).
text	Variable-length non-Unicode data with a maximum length of 2,147,483,647 characters.

Unicode Character Strings

Data Types:

DATA TYPE	Description
nchar	Maximum length of 4,000 characters.(Fixed length Unicode)
nvarchar	Maximum length of 4,000 characters.(Variable length Unicode)
nvarchar(max)	Maximum length of 2 ³¹ characters (SQL Server 2005 only).(Variable length Unicode)
ntext	Maximum length of 1,073,741,823 characters. (Variable length Unicode)

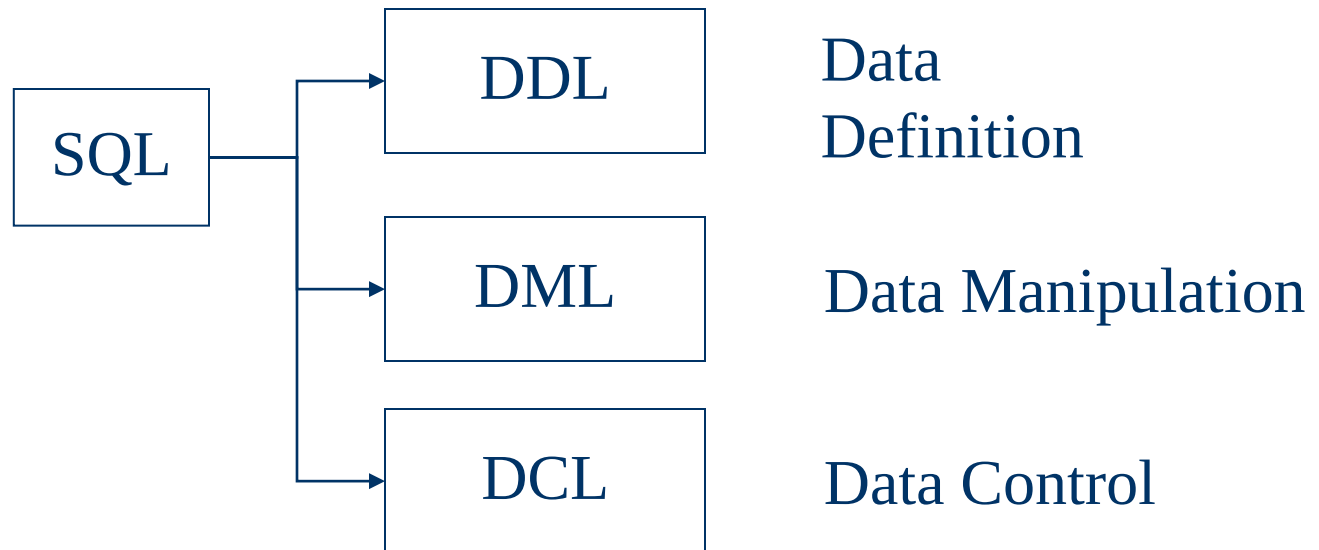
Binary Data Types:

DATA TYPE	Description
binary	Maximum length of 8,000 bytes(Fixed-length binary data)
varbinary	Maximum length of 8,000 bytes.(Variable length binary data)
varbinary(max)	Maximum length of 231 bytes (SQL Server 2005 only). (Variable length Binary data)
image	Maximum length of 2,147,483,647 bytes. (Variable length Binary Data)

Components of SQL – DDL, DML, DCL, DQL.

SQL

- Command line tool that process user's SQL statements
- Requires Oracle account



DDL - Data Definition Language:

Command	Description
CREATE	Creates a new table, a view of a table, or other object in database
ALTER	Modifies an existing database object, such as a table.
DROP	Deletes an entire table, a view of a table or other object in the database.
TRUNCATE	remove all records from a table, including all spaces allocated for the records are removed
COMMENT	add comments to the data dictionary
RENAME	rename an schema object

DML - Data Manipulation Language:

Command	Description
SELECT	Retrieves certain records from one or more tables
INSERT	Insert data into a table
UPDATE	Updates existing data within a table
DELETE	deletes all records from a table, the space for the records remain
CALL	Call a PL/SQL or Java subprogram
EXPLAIN PLAN	explain access path to data
LOCK TABLE	control concurrency

DCL - Data Control Language:

Command	Description
GRANT	Gives a privilege to user
REVOKE	Takes back privileges granted from user

TCL - Transaction Control Language:

Command	Description
COMMIT	save work done
SAVEPOINT	identify a point in a transaction to which you can later roll back
ROLLBACK	restore database to original since the last COMMIT
SET TRANSACTION	Change transaction options like isolation level and what rollback segment to use

SQL Commands

- ◆ Example of SQL stmt: **select * from tab;**
- ◆ **describe <table name>**
- ◆ **list** : list current sql stmt
- ◆ **edit** : edit current sql statement (or ! Editorname <filename>) where editor name is vi etc.
- ◆ **input** : add one or more lines to sqlplus buffer
- ◆ **spool** : start directing output of sql statements to a file
- ◆ **spool off** : turn the spool off
- ◆ **run (or /)** : execute the statement in the current buffer
- ◆ **save <filename.sql>** : save current sql stmt to a file
- ◆ **get <filename.sql>** : load sql statements into buffer
- ◆ **spool <filename>**: send output from sql stmt to a file
- ◆ **start <filename.sql>**: load script file with sql stmts and run them
- ◆ **help**


```
SQL> create table student
2  (name varchar2(10),
3   rollno number(5),
4   age number(3));
```

Typing a SQL Command

Table created.

```
SQL> save c.sql
SP2-0540: File "c.sql" already exists.
Use "SAVE filename[.ext] REPLACE".
```

Saving SQL Command in a file

```
SQL> save c.sql replace
Wrote file c.sql
SQL> ed c.sql
```

Editing SQL Command in a file.

c - Notepad

File Edit Format View Help

```
create table student
(name varchar2(10),
rollno number(5),
age number(3))
/
```

```
2  (name varchar2(10),
3  rollno number(5),
4  age number(3));
```

Table created.

```
SQL> save c.sql
SP2-0540: File "c.sql" already exists.
Use "SAVE filename[.ext] REPLACE".
SQL> save c.sql replace
Wrote file c.sql
SQL> ed c.sql
```

```
SQL> start c.sql
create table student
*
```

ERROR at line 1:
ORA-00955: name is already used by an existing object

```
SQL> ed
Wrote file afiedt.buf
```

```
1  create table student11
2  (name varchar2(10),
3  rollno number(5),
4* age number(3))
```

```
SQL> start c.sql
create table student
*
```

ERROR at line 1:
ORA-00955: name is already used by an existing object

```
SQL> ed c.sql
```

```
SQL> start c.sql
```

Table created.

```
SQL> _
```

Executing the SQL command in a file

SQL Statements

- SELECT
- INSERT
- UPDATE
- DELETE
- MERGE

Data manipulation language (DML)

- CREATE
- ALTER
- DROP
- RENAME
- TRUNCATE
- COMMENT

Data definition language (DDL)

- COMMIT
- ROLLBACK
- SAVEPOINT

Transaction control language (TCL)

- GRANT
- REVOKE

Data control language (DCL)